4190.428 Introduction to Machine Learning

Support Vector Machines
(Part 2)

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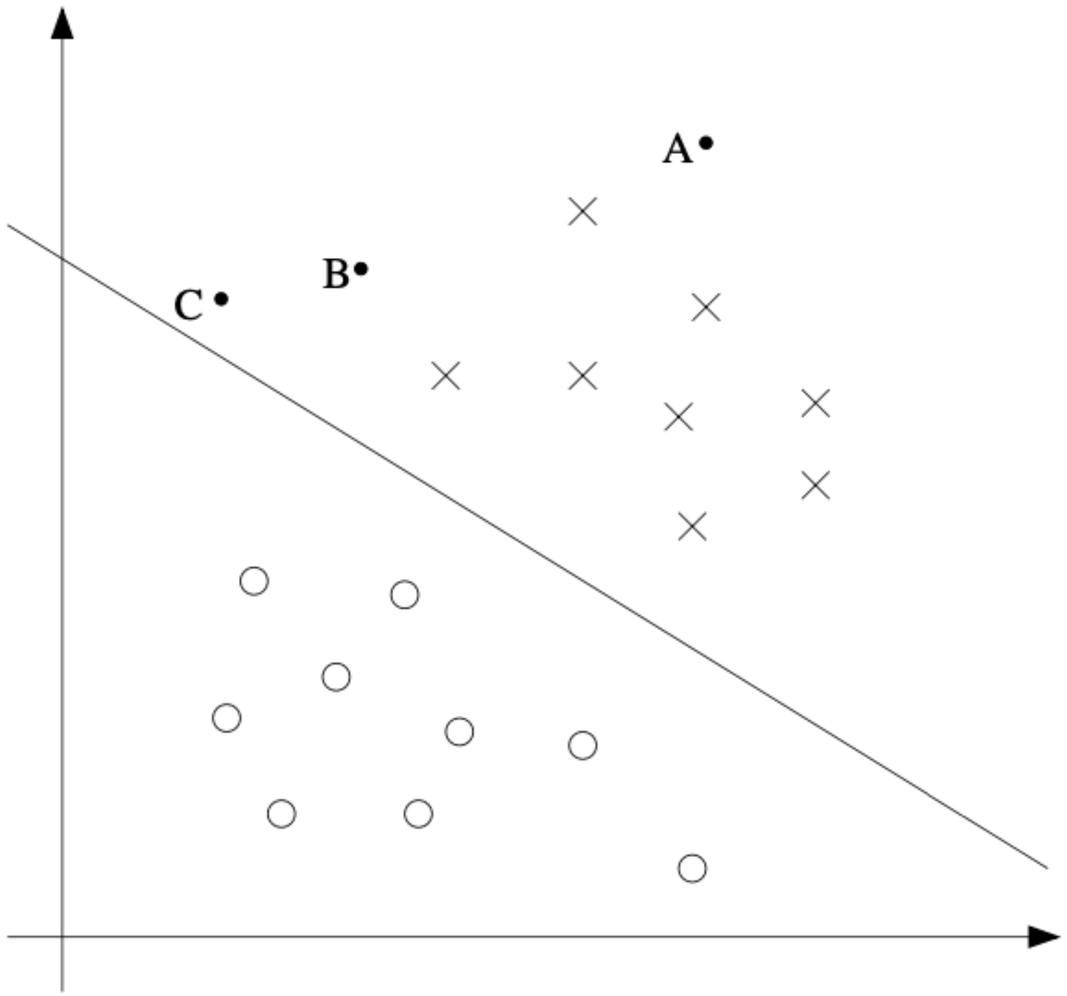
Margins: intuition

- We manage to find a decision boundary that allows us to make all correct and confident predictions on the traing examples



The Optimal Margin Classifier

- The problem of maximize γ :

$$\begin{aligned} \max_{\gamma,w,b} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)}(w^\top x^{(i)} + b) \geq \gamma, \quad i = 1,\ldots,n, \\ & \|w\| = 1 \end{aligned}$$

- $\|w\| = 1$ constraint is non-convex, so solving is difficult.
- We need to transform the problem

The Optimal Margin Classifier

- The problem of maximize γ:

$$\max_{\hat{\gamma},w,b} \quad \frac{\hat{\gamma}}{\|w\|}$$
$$\text{s.t.} \quad y^{(i)}(w^\top x^{(i)} + b) \geq \hat{\gamma}, \quad i = 1,\ldots,n$$

- $\|w\| = 1$ constraint is removed.
- Transform optimization problem, using $\dfrac{\hat{\gamma}}{\|w\|} = \dfrac{1}{\|w\|}$

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y^{(i)}(w^\top x^{(i)} + b) \geq 1, \quad i = 1,\ldots,n$$

- Its solution gives us the **optimal margin classifier**
- This optimization problem can be solved using commercial **quadratic programming** (QP) code —> any other solution?

Backgrounds: Lagrange Multipliers

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- Consider a problem of the following form:

$$\min_{w} f(w) \quad \text{s.t.} \quad h_i(w) = 0, \quad i = 1, \ldots, l$$

- Define **Lagrangian** : $\mathscr{L}(w, \beta) = f(w) + \sum_{i=1}^{l} \beta_i h_i(w)$
- β_i's are called **Lagrange multipliers**. We would then find and set $\mathscr{L}$'s partial derivates to zero:

$$\frac{\partial \mathscr{L}}{\partial w_i} = 0, \quad \frac{\partial \mathscr{L}}{\partial \beta_i} = 0$$

- and solve for w and β

Backgrounds: Generalized Lagrangian

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- Let's generalize this to **constrained optimization** problems
  - The problems has **inequality** as well as **equality constraints**
- **Primal** optimization problem:

$$\begin{aligned} \min_{w} \quad & f(w) \\ \text{s.t.} \quad & g_i(w) \leq 0, \quad i = 1, \ldots, k, \\ & h_i(w) = 0, \quad i = 1, \ldots, l \end{aligned}$$

- **Generalized Lagrangian**: $\mathscr{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) + \sum_{i=1}^{l} \beta_i h_i(w)$ [Eq. Lag]
- Here, the α's and β's are the Lagrange multipliers.
- Let's consider the quantity:

$$\theta_{\mathscr{P}}(w) = \max_{\alpha, \beta : \alpha_i \geq 0} \mathscr{L}(w, \alpha, \beta)$$

Backgrounds: Primal Constraints

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- Here, the "$\mathscr{P}$" subscript stands for "**primal**".
- Let some w be given. If w violates any of the **primal constraints** (either $g_i(w) > 0$ or $h_i(w) \neq 0$ for some i), then it should be able to verify that

$$\theta_{\mathscr{P}}(w) = \max_{\alpha,\beta:\alpha_i \geq 0} \left(f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) + \sum_{i=1}^{l} \beta_i h_i(w) \right)$$
$$= \infty \, .$$

- Conversely, if the constraints are indeed satisfied for a particular value of w, then $\theta_{\mathscr{P}}(w) = f(w)$. Hence,

$$\theta_{\mathscr{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise} \end{cases}$$

Backgrounds: Primal Constraints

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- $\theta_{\mathscr{P}}$ takes the same value as the objective in our problem for all values of w that satisfies the primal constraints, and is positive infinity if the constraints are violated.

- If we consider the minimization problem:

$$\min_{w} \theta_{\mathscr{P}}(w) = \min_{w} \max_{\alpha,\beta:\alpha_i \geq 0} \mathscr{L}(w, \alpha, \beta)$$

- We see that it is the same problem (has the same solution as) our original, primal problem.
- For later use, we define **value** of the primal problem:

$$p^* = \min_{w} \theta_{\mathscr{P}}(w)$$

Backgrounds: Dual Problem

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- Let's define another quantity:

$$\theta_{\mathscr{D}}(\alpha, \beta) = \min_{w} \mathscr{L}(w, \alpha, \beta)$$

- Where $\mathscr{D}$ stands for '**dual**'.
- Pose the **dual** optimization problem:

$$\max_{\alpha,\beta:\alpha_i \geq 0} \theta_{\mathscr{D}}(\alpha, \beta) = \max_{\alpha,\beta:\alpha_i \geq 0} \min_{w} \mathscr{L}(w, \alpha, \beta)$$

- That is exactly the same as our primal problem shown before, except that the **order** of the "max" and the "min" are now **exchanged**.

Backgrounds: Lagrange Duality

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- We also define the optimal value of the dual problem's objective to be $d^* = \max_{\alpha,\beta:\alpha_i \geq 0} \theta_{\mathscr{D}}(w)$, then we have:

$$d^* = \max_{\alpha,\beta:\alpha_i \geq 0} \min_{w} \mathscr{L}(w, \alpha, \beta) \leq \min_{w} \max_{\alpha,\beta:\alpha_i \geq 0} \mathscr{L}(w, \alpha, \beta) = p^*$$

- This follows from the "max min" of a function always being less than or equal to the "min max."
- Under certain conditions, we will have

$$d^* = p^*$$

Backgrounds: Lagrange Duality

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- Suppose
  - (1) f and g_i are convex
  - (2) h_i are affine (there exists a_i, b_i, so that $h_i(w) = a_i^\top w + b_i$)
  - (3) g_i are strictly feasible (there exists w so that $g_i(w) < 0$ for all i)

- Then, there must exist w^*, α^*, and β^* so that
  - w^* is the solution to the **primal problem**
  - α^* and β^* are the solution to the **dual problem.**
- Moreover, $p^* = d^* = \mathscr{L}(w^*, \alpha^*, \beta^*)$

Backgrounds: KKT conditions

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- Under the previous assumptions, w^*, α^*, and β^* satisfy the **Karush-Kuhn-Tucker (KKT) conditions**, which are as follows:

$$\frac{\partial}{\partial w_i}\mathscr{L}(w^*, \alpha^*, \beta^*) = 0, \quad i = 1, \ldots, d$$

$$\frac{\partial}{\partial \beta_i}\mathscr{L}(w^*, \alpha^*, \beta^*) = 0, \quad i = 1, \ldots, l$$

$$\alpha_i^* g_i(w^*) = 0, \quad i = 1, \ldots, k$$

KKT dual complementarity condition

$$g_i(w^*) \leq 0, \quad i = 1, \ldots, k$$

$$\alpha^* \geq 0, \quad i = 1, \ldots, k$$

Backgrounds: KKT dual complementarity condition

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- **KKT dual complementarity** condition:

$$\alpha_i^* g_i(w^*) = 0, \quad i = 1, \ldots, k$$

- It implies that if $\alpha_i^* > 0$ then $g_i(w^*) = 0$.
  - i.e., $g_i(w) \leq 0$ constraint is **active**
  - it holds with equality rather than with inequality
- This will be key for showing that the SVM has only a **small number of support vectors**

Optimal Margin Classifiers: the Dual Form

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- We posed the following (primal) optimization problem for finding the optimal margin classifier :

$$\min_{w,b} \quad \frac{1}{2}\|w\|^2$$
$$\text{s.t.} \quad y^{(i)}\left(w^\top x^{(i)} + b\right) \geq 1, \quad i = 1,\ldots,n$$

- We can write the constraints as

$$g_i(w) = -\,y^{(i)}\left(w^T x^{(i)} + b\right) + 1 \leq 0$$

Optimal Margin Classifiers

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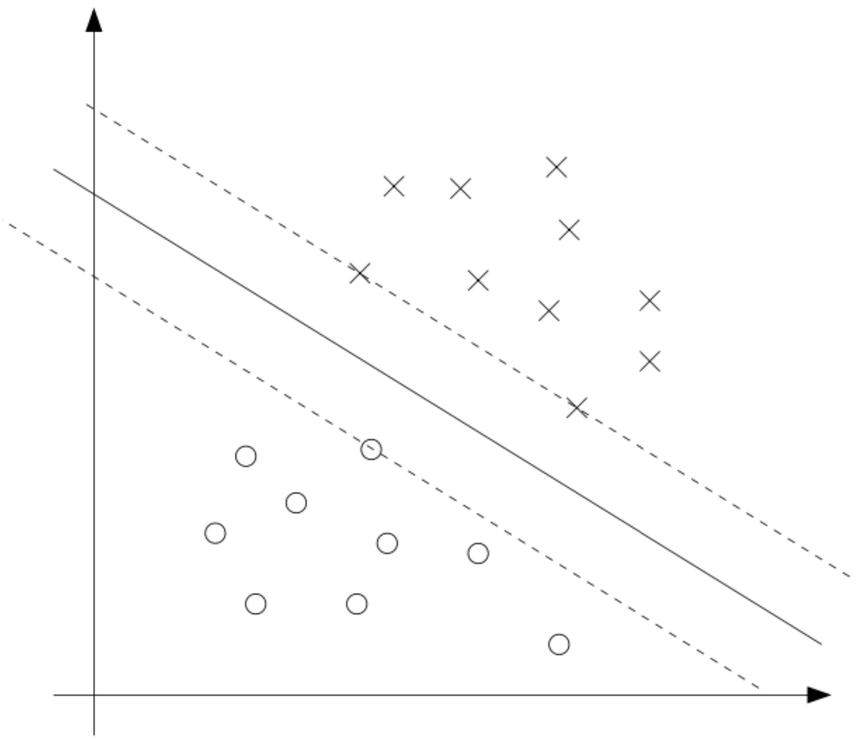
- In the right figure, a maximum margin separating hyperplane is shown by the solid line
- From the KKT dual complementarity condition, we will have $\alpha_i > 0$ only for the training examples that have functional margin exactly equal to one (the ones hold with equality $g_i(w) = 0$)
- The three points on the dashed lines are called **support vectors** in this problem



Optimal Margin Classifiers: the Dual Form

ADVANCED

- When we construct the Lagrangian for our optimization problem we have:

$$\mathscr{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{n} \alpha_i \left[y^{(i)}\left(w^\top x^{(i)} + b\right) - 1\right]$$

- Let's find the dual form of the problem.

$$\nabla_w \mathscr{L}(w, b, \alpha) = w - \sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)} = 0$$

Optimal Margin Classifiers: the Dual Form

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- This implies that

$$w = \sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)} . \text{ [Eq. w]}$$

- As for the derivative with respect to b, we obtain

$$\frac{\partial}{\partial b} \mathscr{L}(w, b, \alpha) = \sum_{i=1}^{n} \alpha_i y^{(i)} = 0. \text{ [Eq. b]}$$

Optimal Margin Classifiers: the Dual Form

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- If we take [Eq. w] and plug that back into the Lagrangian, and simplify, we get:

$$\mathscr{L}(w, b, \alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^\top x^{(j)} - b \sum_{i=1}^{n} \alpha_i y^{(i)} .$$

- But the last term must be zero by [Eq. b], so we obtain:

$$\mathscr{L}(w, b, \alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^\top x^{(j)} .$$

- Recall that we got to the equation above by minimizing $\mathscr{L}$ w.r.t. w and b.

Optimal Margin Classifiers: the Dual Form

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- Putting this together with the constraints $\alpha_i \geq 0$ and the constraint, we obtain the following **dual optimization problem**:

$$\max_{\alpha} \quad W(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle,$$

$$\text{s.t.} \quad \alpha_i \geq 0, \quad i = 1, \ldots, n,$$

$$\sum_{i=1}^{n} \alpha_i y^{(i)} = 0.$$

Optimal Margin Classifiers: the Dual Form

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- You should also be able to verify that the conditions required for $p^* = d^*$ and the KKT conditions to hold are indeed satisfied in our optimization problem.
- Having found w^*, by considering the primal problem, it is also straightforward to find the optimal value for the intercept term b as [Try yourself]:

$$b^* = -\frac{\max_{i:y^{(i)}=-1} w^\top x^{(i)} + \min_{i:y^{(i)}=1} w^\top x^{(i)}}{2}.$$

Optimal Margin Classifiers: the Dual Form

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- We would calculate $w^\top x + b$:

$$w^\top x + b = \left(\sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)} \right)^\top x + b$$
$$= \sum_{i=1}^{n} \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b \,.$$

- The resulting algorithm, **support vector machines**, will be able to efficiently learn in very high dimensional spaces.

Regularization and the Non-separable Case

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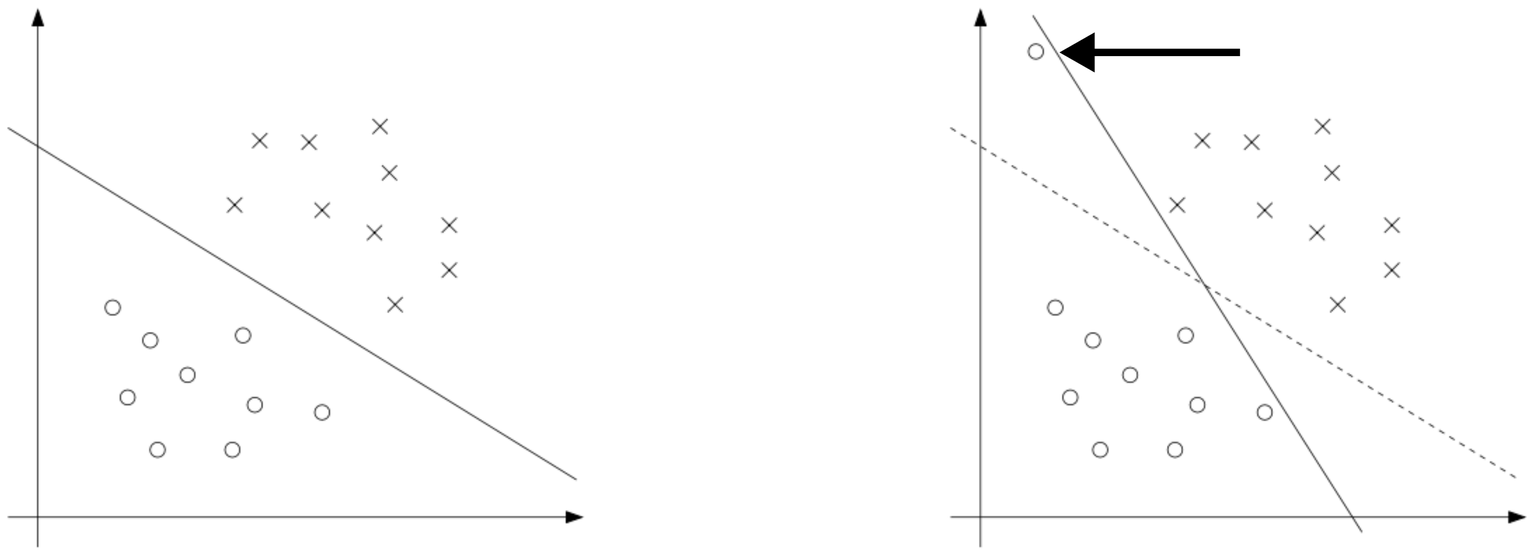

- [Left] Optimal margin classifier
- [Right] A single outlier added in the upper-left region causes the decision boundary to make a dramatic swing, resulting classifier has a much smaller margin
- **Question**: can we make the algorithm work for non-linearly separable datasets and be less sensitive to outliers?

Regularization and the Non-separable Case

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- We reformulate our optimization as follows:

$$\min_{w,b,\xi} \quad \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i$$

$$\text{s.t.} \quad y^{(i)}\left(w^\top x^{(i)} + b\right) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1,\ldots,n\,.$$

- Examples are now **permitted to have (functional) margin less than 1** ($1 - \xi_i$, with $\xi_i > 0$), the objective function is increased by $C\xi_i$.
- The parameter C controls the relative weighting between $\|w\|^2$ small while ensuring that most examples have functional margin at least 1.

Regularization and the Non-separable Case

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- Like before, we can form the Lagrangian [Eq. Lag]:

$$\mathscr{L}(w, b, \xi, \alpha, r) = \frac{1}{2} w^\top w + C \sum_{i=1}^{n} \xi_i - \sum_{i=1}^{n} \alpha_i \left[y^{(i)} (x^{(i)\top} w + b) - 1 + \xi_i \right] - \sum_{i=1}^{n} r_i \xi_i \,.$$

- After setting the derivatives with respect to w and b to zero as before, substituting them back in, and simplifying, we obtain the following dual form of the problem [Try yourself]:

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle, \\ \text{s.t.} \quad & 0 \le \alpha_i \le C, \quad i = 1, \ldots, n, \\ & \sum_{i=1}^{n} \alpha_i y^{(i)} = 0. \end{aligned}$$

Regularization and the Non-separable Case

ADVANCED

- Also, the KKT dual-complementarity conditions are:

$$\alpha_i = 0 \implies y^{(i)}\left(w^\top x^{(i)} + b\right) \geq 1,$$

$$\alpha_i = C \implies y^{(i)}\left(w^\top x^{(i)} + b\right) \leq 1,$$

$$0 < \alpha_i < C \implies y^{(i)}\left(w^\top x^{(i)} + b\right) = 1.$$

- Now all that remains is to give an algorithm for actually solving the dual problem —> SMO algorithm!
- KKT dual-complementarity conditions are useful for testing for the convergence of the SMO algorithm

Coordinate Ascent Algorithm

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- Consider trying to solve the unconstrained optimization problem:

$$\max_{\alpha} W(\alpha_1, \alpha_2, \ldots, \alpha_n) .$$

- W as just some function of the parameters α_i
- The new algorithm we're going to consider here is called **coordinate ascent**:

Loop until convergence {

For $i = 1, \ldots, n$, {

$$\alpha_i := \arg\max_{\hat{\alpha}_i} W(\alpha_1, \ldots, \alpha_{i-1}, \hat{\alpha}_i, \alpha_{i+1}, \ldots, \alpha_n) .$$

}

}

Coordinate Ascent Algorithm

ADVANCED

- Innermost of loop of this algorithm holds all the variables except for some α_i fixed, and reoptimize W with respect to just the parameter α_i.
- A more sophisticated version might choose other orderings that can make the largest increase in $W(\alpha)$

Loop until convergence {

For $i = 1, \ldots, n$, {

$$\alpha_i := \arg\max_{\hat{\alpha}_i} W(\alpha_1, \ldots, \alpha_{i-1}, \hat{\alpha}_i, \alpha_{i+1}, \ldots, \alpha_n) .$$

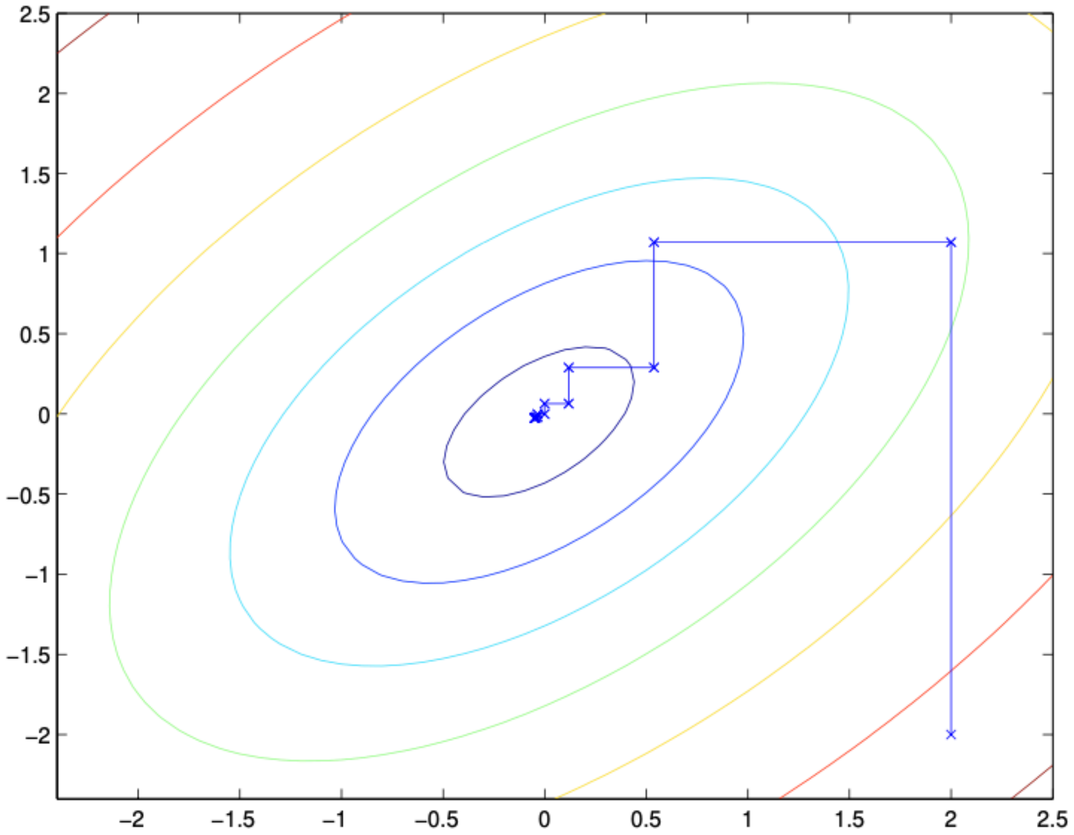
}

}

Coordinate Ascent Algorithm

ADVANCED

- The ellipses in the figure are the contours of a quadratic function that we want to optimize.
- Coordinate ascent was initialized at (2,-2), and also plotted in the figure is the path that it took on its way to the global maximum.
- Takes a step that's parallel to one of the axes, since only one variable is being optimized



The SMO Algorithm

ADVANCED

- We close off the discussion of SVMs by sketching the derivation of the SMO algorithm.
- Here's the (dual) optimization problem that we want to solve :

$$\max_{\alpha} \quad W(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle,$$

$$\text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad i = 1, \ldots, n,$$

$$\sum_{i=1}^{n} \alpha_i y^{(i)} = 0.$$

- The constraint ensures that: $\alpha_1 y^{(1)} = -\sum_{i=2}^{n} \alpha_i y^{(i)}$.
- Or, by multiplying both sides by $y^{(1)}$, we equivalently have: $\alpha_1 = -y^{(1)} \sum_{i=2}^{n} \alpha_i y^{(i)}$.

The SMO Algorithm

ADVANCED

- Let's say we currently have some setting of the α_i's that satisfy the constraints, and suppose we've decided to hold $\alpha_3, \ldots, \alpha_n$ fixed, and want to re-optimize $W(\alpha_1, \alpha_2, \ldots, \alpha_n)$ with respect to α_1 and α_2.
- We require that :

$$\alpha_1 y^{(1)} + \alpha_2 y^{(2)} = -\sum_{i=3}^{n} \alpha_i y^{(i)} .$$

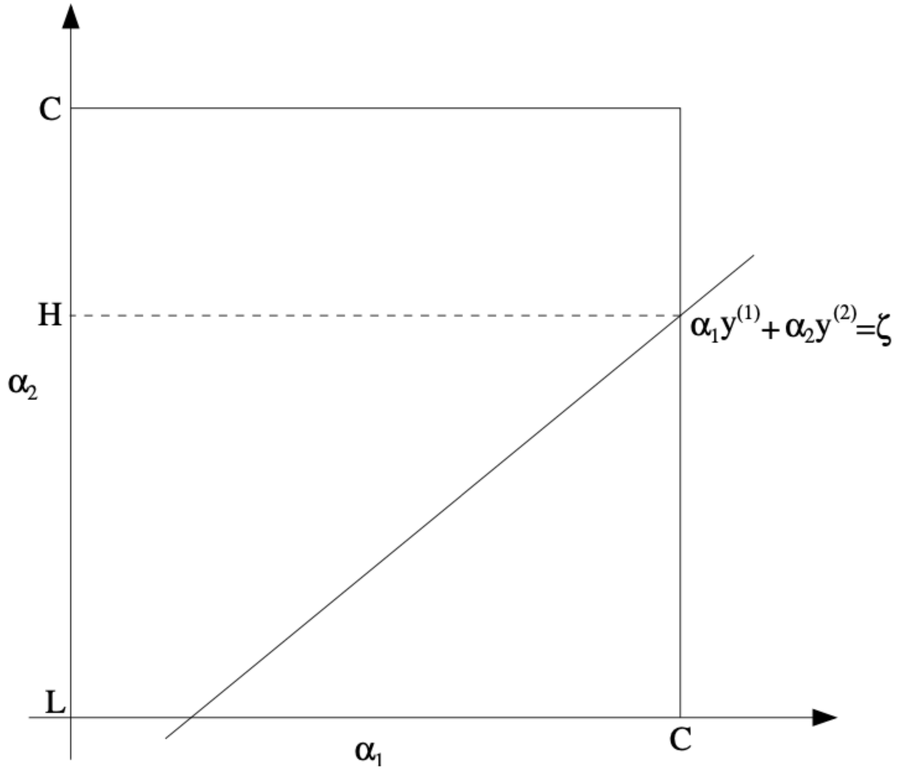
- Since the right hand side is fixed, we can just let it be denoted by some constraint ζ (zeta):

$$\alpha_1 y^{(1)} + \alpha_2 y^{(2)} = \zeta .$$

The SMO Algorithm

ADVANCED

- From the constraints, we know that α_1 and α_2 must lie within the box $[0,C] \times [0,C]$ shown.
- Also plotted is the line $\alpha_1 y^{(1)} + \alpha_2 y^{(2)} = \zeta$, on which we know α_1 and α_2 must lie.
- We know $L \leq \alpha_2 \leq H$
- The will be some lower-bound L and some upper-bound H on the permissible values for α_2 that will ensure that α_1, α_2 lie within the box $[0,C] \times [0,C]$



The SMO Algorithm

ADVANCED

- We can write α_1 as function of α_2 :

$$\alpha_1 = (\zeta - \alpha_2 y^{(2)}) y^{(1)} .$$

- Hence, the objective $W(\alpha)$ can be written :

$$W(\alpha_1, \alpha_2, \ldots, \alpha_n) = W((\zeta - \alpha_2 y^{(2)}) y^{(1)}, \alpha_2, \ldots, \alpha_n) .$$

- Treating $\alpha_3, \ldots, \alpha_n$ as constants, you should be able to verify that this is just some quadratic function in α_2.

The SMO Algorithm

ADVANCED

- You should also be able to convince yourself that if we had instead wanted to maximize W with respect to α_2 but subject to the box constraint, then we can find the resulting value optimal simply by taking $\alpha_2^{\text{new, unclipped}}$ and "clipping" it to lie in the $[L, H]$ interval, to get

$$\alpha_2^{\text{new}} = \begin{cases} H & \text{if } \alpha_2^{\text{new, unclipped}} > H, \\ \alpha_2^{\text{new, unclipped}} & \text{if } L \leq \alpha_2^{\text{new, unclipped}} \leq H, \\ L & \text{if } \alpha_2^{\text{new, unclipped}} < L\,. \end{cases}$$

QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)